#manydaughters – Hypotheses
research team ID:

pid – Never Changing Person ID
cid – Original Household Number, Case ID
pgpartnr – Partner person number
The dataset is limited to the years 2000 to 2020 and includes all individuals who have completed the individual questionnaire in at least one of these years (n = 90,357)

Note on robustness checks

Our analysis focuses on the effect of having "at least one daughter" on the support for gender equality or the political preferences of respondents. To our understanding there should only be a regression equation for each hypothesis. However, we could consider robustness checks, as follows:

(i) examine subsample of respondents with kids under 11 years old, or include dummy variables for daughters under 5, between 6 and 10, or older than 11 years old.
(ii) binarize the dependent variable; e.g., we could define "low-support" and "high-support" for gender equality in H1, or "left-wing" and "right-wing" voters in H2.
(iii) control for the family size or the number of daughters.

We avoid these robustness checks in our pre-analysis plan to focus on the effect of having "at least one daughter" and the requirement of "one table/regression for hypothesis". If needed, we would be happy to provide these additional regressions.

H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes).*

That is, denoting X as a measure of support for gender equality,
(X for respondents with daughters) > (X for respondents without daughters).

Dependent variable: In the topics of SOEP-Core, there is one specific topic on *Attitudes, Values, and Personality*. And specifically, there is one module on *Attitudes towards genders* that was administered only in 2019. This module, plh0395, aks to what extent the respondent agrees with a number (six) of statements. The answers range from 1 (do not agree at all) to 7 (completely agree). (https://paneldata.org/soep-core/datasets/pl/plh0395i05).

In our opinion, the question that best captures gender equality is the fifth one: "*Schools and preschools should promote non-gender-typical activities that are not typical for children's gender*". However, we will construct a measure that considers the average answer to all questions -this is to include issues of gender equality, identity, and diversity.

Exclusion criteria: We restrict attention to respondents with kids in 2019 (using the info in BIOBIRTH/biokids) to avoid selection bias.

Independent variables:
D: Dummy daughter: it takes value 1 if the respondent has at least one daughter (using the info in BIOBIRTH/kidsex…)

OLS (Robust standard errors)

$$plh0395i05_i = \alpha + \beta_1 D_i + controls \qquad (1)$$

No daughters -> α
At least a daughter -> $\alpha + \beta_1$

$$H_1: \beta_1 > 0$$

Controls:

- Gender (BIOBIRTH/sex) -dummy Female-
- Age and age_squared/100 – (BIOBIRTH/Year of birth)
- Education: highest degree obtained (PGEN/pgisced11) – dummies primary, secondary, tertiary, college (bachelor, master, doctoral), other.
- Marital status (PGEN/pgfamstd) – dummies single, married, separated, divorced, widow.
- Employment status (PGEN/pgemplst) – dummies full time, part-time, vocational training, not-employed, other.
- Income: (PGEN/pglabnet) – natural logarithm of current net labour income
- Religion (PL/plh0258_h) – dummies Catholic, Islamic, other religion, No
- East Germany (HGEN/hgnuts1_ew) – dummy 1 if living in East Germany
- Emigrant (PGEN/pgnation) – dummy non-german

H2. *Having daughters leads to more left-leaning political preferences*.

That is, denoting Y as a measure of left-leaning political preferences,
(Y for respondents with daughters) > (Y for respondents without daughters).

Dependent variable: plh0004 (In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours? It ranges from 0 -extreme left- to 10 -extreme right-). (https://paneldata.org/soep-core/datasets/pl/plh0004). It is available for periods t = 2005, 2009, 2014 and 2019. (A panel :-)

Exclusion criteria: We restrict our attention to respondents with kids at some point in the panel (to avoid selection bias). We then define $C_{i,t}$ = 1 if individual *i* has a kid at time t (otherwise $C_{i,t} = 0$) and exclude individuals where $\sum_t C_{i,t} = 0$

Independent variables:
$D_{i,t}$: Dummy daughter: it takes value 1 if the respondent has at least one daughter at time t.

Panel. Xtreg with fixed effects at pid level.

$$plh0004_{i,t} = \alpha + \beta_1 D_{i,t} + controls$$

Test:

$$H_1: \beta_1 > 0$$

Controls:

- Gender (BIOBIRTH/sex) -dummy Female-
- Age and age_squared/100 – (BIOBIRTH/Year of birth)
- Education: highest degree obtained (PGEN/pgisced11) – dummies primary, secondary, tertiary, college (bachelor, master, doctoral), other.
- Marital status (PGEN/pgfamstd) – dummies single, married, separated, divorced, widow.
- Employment status (PGEN/pgemplst) – dummies full time, part-time, vocational training, not-employed, other.
- Income: (PGEN/pglabnet) – natural logarithm of current net labour income
- Religion (PL/plh0258_h) – dummies Catholic, Islamic, other religion, No
- East Germany (HGEN/hgnuts1_ew) – dummy 1 if living in East Germany
- Emigrant (PGEN/pgnation) – dummy non-german
- Year dummies

H3. *Having daughters leads to more gender equality in the household (as measured by behaviours or outcomes).*

Dependent variable: pli0016_h (Now, we'd like to know how your normal everyday life currently looks. For the following activities, please tell us *how many hours you spend on them on an average weekday, on a typical Saturday and on a typical Sunday. Housework such as washing, cooking, cleaning*).

(https://paneldata.org/soep-core/datasets/pl/pli0016_h) The variable takes different labels, see description: Step 1: The variable pli0016_h is replaced by the variable pli0016_v1 for the years 1984 to 1990. Step 2: The variable pli0016_h is replaced by the variable pli0016_v4….

Dependent variable: We need to construct a variable that stands for the percentage of housework by men in the household.

$$PHmen_{i,t} = \frac{\text{Hours of housework by men}}{\text{Hours of housework by men + Hours of housework by women}} \text{ x } 100$$

Exclusion criteria: We restrict our attention to (i) households with same (heterosexual) partners (men/women) living together and (ii) respondents with at least one kid during the panel (we will use the same procedures as in H2 to exclude respondents who do not satisfy these criteria). Years 2009, 2011, 2013, 2015, 2017 and 2019.

Independent variable:
That is, denoting Z as a measure of gender equality in the household,
(Z for respondents with daughters) > (Z for respondents without daughters).

Panel . xtreg fixed effects at cid level.

$$PHmen_{i,t} = \alpha + \beta_1 D_{i,t} + controls$$

Test:

$$H_1: \beta_1 > 0$$

Controls:

- Age and Age_sq/100 of husband (BIOBIRTH/Year of birth)
- Age and Age_sq/100 of wife (BIOBIRTH/Year of birth)
- Level of education of husband (PGEN/pgisced11)
- Level of education of wife (PGEN/pgisced11)
- Percentage of net labour income by husband (PGEN/pglabnet)
- Employment status of husband (dummy = 1 if full-time) (PGEN/pgemplst)
- Employment status of wife (dummy = 1 if full-time) (PGEN/pgemplst)
- Year dummies
- Total number of hours devoted to housework by the household (PL/pli0016_h)

H4. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*

That is, denoting X as a measure of support for gender equality,
(X for male respondents with daughters − X for male respondents without daughters) > (X for female respondents with daughters − X for female respondents without daughters)

Our analysis follows the same logic as in H1 but we include the interaction between the treatment variable (D_i) and the responder's gender ($Female_i$)

OLS

$$plh0395i05_i = \alpha + \beta_1 D_i + \beta_2 Female + \beta_3 D_i Female_i + controls \qquad (1)$$

$$H_1: \beta_3 < 0$$

Controls.

- Gender (BIOBIRTH/sex) -dummy Female-
- Age and age_squared/100 – (BIOBIRTH/Year of birth)
- Education: highest degree obtained (PGEN/pgisced11) – dummies primary, secondary, tertiary, college (bachelor, master, doctoral), other.
- Marital status (PGEN/pgfamstd) – dummies single, married, separated, divorced, widow.
- Employment status (PGEN/pgemplst) – dummies full time, part-time, vocational training, not-employed, other.
- Income: (PGEN/pglabnet) – natural logarithm of current net labour income
- Religion (PL/plh0258_h) – dummies Catholic, Islamic, other religion, No
- East Germany (HGEN/hgnuts1_ew) – dummy 1 if living in East Germany
- Emigrant (PGEN/pgnation) – dummy non-german